

The Mohr Method

Lishan Prado

June 2026

1 The Mohr Method

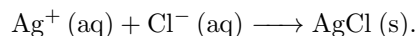
The Mohr method is a type of titration used to determine the amount of chloride ions in a solution. We begin by adding silver nitrate to a solution of chloride ions, causing silver chloride to precipitate.



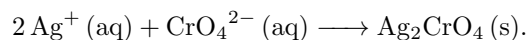
Figure 1: Silver Chloride

Figure 1 shows a picture of the silver chloride precipitate, which is milky white. The picture was taken from the Science Photo Library.

Once all of the chloride has been precipitated, any excess silver ions added will react with the chromate indicator to form a brick-red silver chromate precipitate, signaling the endpoint of the titration.



The above reaction is very favorable, with K_{sp} of AgCl equal to $1.8 \cdot 10^{-10}$.



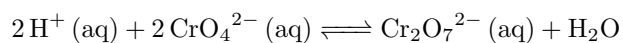
The reaction above, while also favorable, is not as favorable as that of the formation of silver chloride. This is why silver chromate will not form until enough chloride has been used up.



Figure 2: Silver Chromate

The figure above shows a picture of silver chromate precipitate. It may be interesting to note that the pipette above the solution is yellow. This is because the color of chromate is yellow.

One cool equilibrium that is slightly related is that between chromate and dichromate:



The color of the chromate ion is famously yellow, while dichromate is orange.



Figure 3: Dichromate and Chromate, respectively

An interesting problem about Mohr's method comes from problem 3 of Round 2 of ACOT 2025. Here's a link.

If you have any questions or comments, feel free to contact me! Check out my website for more chemistry handouts that I've written.